THE EFFECT OF WEATHER ON SPORT

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IN September 1975 *Crossbow* became the first sail boat in the world to break sailing's equivalent of the 'four minute mile' – the 30 kt barrier.* She registered 31·09 kt beating her own previous world record speed of 29·3 kt. However the press release which shortly followed did not mention that the average wind speed for the previous record was 20·0 kt, whereas the average for the new record was 25·2 kt! How significant was the 25 per cent increase in wind speed?

A few years ago, on the continent, the ladies world long-jump record was broken by a girl who finished fourth in an event. Neither her attempt, nor those of the first three, counted: the wind assistance was considered to be too great.

In the 1974 county cricket championship Hampshire looked set to take the title and £3000 prize money until their final fixture against Yorkshire was 'washed out' without a ball being bowled. In fact they lost 5 days play from their last three matches due to rain, allowing Worcestershire to take the title by 2 points.

These three examples serve to illustrate the diverse effects of weather on sport. In the first the weather, in the form of wind, probably helped to break the world sailing speed record. In the second example the wind denied potential world records in the long jump. In the third the weather assumed its more familiar role as an interference factor. This paper attempts to classify similar interactions between weather and sport. First, in order to limit the discussion, we must define the scope of the term sport. Gillett (1971) gives a useful insight: 'Sport is not a (board) game, or a pastime, or a contest between machines, it is a contest between at least two people, involving their brains and bodies, with each group or individual trying to beat the other. So discussion of sport will not include cards, chess, mountain climbing, horse racing or fishing which are either non-physical or not really competitive'. This is too stringent a definition for the purpose of this article which will also include sports that utilise atmospheric energy such as gliding, but it does allow us to exclude horse racing and motor racing, both of which rely on energy other than that provided by the human body or by the weather.

* The record was achieved during the John Player World Sailing Speed event at Weymouth on 30 September 1975, organised by the Royal Yachting Association. For a discussion of the history of the event and rules of contest, consult the RYA's press release entitled 'In search of speed' which begins with the words: 'The action of wind on sail is the most primitive form of propulsion, but to achieve optimum speed calls for exceptional skill and knowledge of the elements . . .'

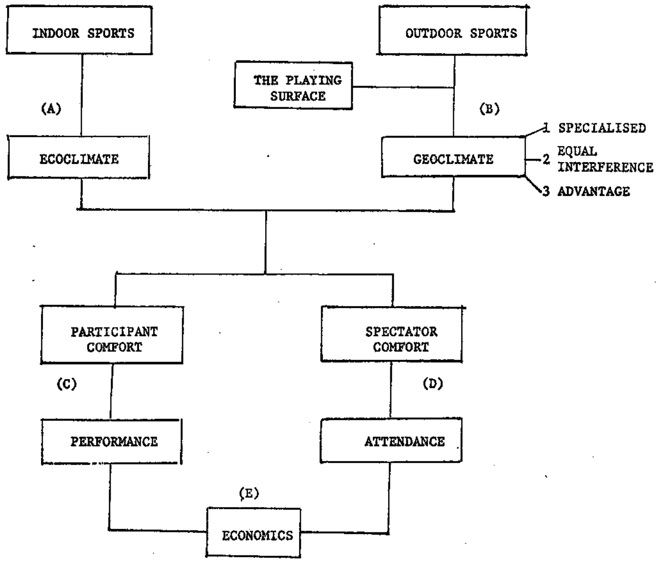

Fig. 1. The effect of weather on sport

Nevertheless both are very sensitive to weather. The 1975 British Grand Prix was abandoned when cars with dry-weather tyres hit a wet section of the track following a heavy shower of rain. The multiple crash which resulted emphasised the problems faced when racing cars encounter short-period weather changes. Horse-race gambling is often governed by the punter's perception of a horse's likely performance on different ground conditions ('the going'), which are principally a function of weather, notably precipitation intensity, duration and timing.

Maunder (1970) classified the effect of weather on sport in two categories – the effect on the sport itself, and the effect upon the spectator. It seems desirable, however, to modify this classification so that the effect of weather on participant comfort is included, and to distinguish between indoor and outdoor sports. Fig. 1 outlines the basic interactions between weather and sport.

INDOOR SPORTS

Sports such as squash or badminton are normally, in this country, played indoors in artificial 'ecoclimates'* where meteorological hinderances such as rain and wind are avoided. There has been a recent increase in the provision

* *Ecoclimate* is taken in this paper to represent the climate and weather of the buildings in which man works, lives and plays sport. The *geoclimate* is taken as the outside climate and weather which is normally measured by standard meteorological methods. Thus the ecoclimate is normally concerned only with temperature, humidity and radiation. These concepts are adapted from the paper by Bates, M., 'The Role of weather in human behaviour' in: Sewell, W. R. D. ed., 1966, *Human dimensions of weather modification*, University of Chicago, Department of Geography, Research Paper No. 105, pp. 393–407

of indoor sports facilities such as sports halls and squash courts. A report by the British National Playing Fields Association (1971) states:

'Our uncertain climate, and the fact that for eight months of the year our evenings are spent in darkness would seem to be ample justification for redressing the present imbalance between outdoor and indoor games provision. From a land-use point of view . . . only America has a greater allocation of outdoor play space than in our country, and west Germany . . . to take it to the other extreme, has almost as much recreation space under cover as that provided outdoors.

We are therefore likely to see increased provision of indoor facilities in the future. Although the worst 'interference weather' elements are removed when considering indoor ecoclimates, nevertheless air temperature and humidity are still very important variables affecting indoor sports. For instance, in the case of badminton, quoting from the handbook *Know the game badminton* (1972) – 'Because of the different atmospheric conditions of different halls and weather, shuttles are made in varying flights . . . in a very clear atmosphere a shuttle will travel further than on a misty day, and likewise the size and temperature of the hall will have some effect upon its behaviour.'

This typifies the subjective description that can be found relating most sports and weather. One might think that the choice of shuttle used should depend upon measurement of the air temperature and humidity relating to air density. Not a bit of it, Law 4 states:

'A shuttle shall be deemed to be of the correct pace if, when a player of average strength strikes it with a full underhand stroke from a spot immediately above the back boundary line in a line parallel to the side lines, and at an upward angle, it falls not less than 0·30 m and not more than 0·76 m short of the other back boundary line'!

This empirical approach underlies the lack of physical understanding of the aerodynamics of shuttles, and this is paralleled in most other indoor and outdoor ball games. It means that in varying conditions the experienced player has a distinct advantage, though it may also be said that it adds variety to the sport.

Table 1 lists the temperature requirements of various sports as given by the National Playing Field Association (1971). These temperatures are for sports halls in which the ecoclimate can be controlled. The ideal temperature for a sport depends on up to three factors, firstly participant comfort, secondly spectator comfort and thirdly, if a ball or other projectile is used, any variation of flight or bounce with temperature. The relative importance of these three factors depends upon the sport. For instance the recommended temperature for badminton is 7°C for tournaments, reflecting the supposed ideal shuttle dynamics, whereas the relatively high temperature recommended for boxing shows a concession to the spectator. Note that there is no recommendation regarding humidity for indoor halls, although the temperature gradient is considered in a different part of the recommendations. 'There

TABLE 1. Recommended temperatures for various indoor sports for community sports halls; from National Playing Fields Association (1971)

Sport	Temperature °C	Sport	Temperature °C
Archery	13–16	Indoor tennis	13–16
Badminton	7 tournament	Judo	13–16
	13 otherwise	Netball	13–16
Basketball	10–13	Rowing tank	13–16
Boxing	13–16	Shooting	13–16
Indoor bowls	16–19	Squash	10–13

should not be more than a 5·5 deg C difference between shoulder height and feet' according to the National Playing Fields Association (1971), though it omits to mention why!

One major advantage of the indoor sport is that the state of the floor is uniform, whereas the state of the ground in outdoor games can lead to very different playing conditions which affect the bounce of the ball and the balance and control of participants.

OUTDOOR SPORTS

The interaction between weather and outdoor sport is more complicated in that we have to consider the 'geoclimate'* in which more meteorological variables are concerned and which we cannot control. Not only are temperature and humidity important, but also wind velocity, visibility, precipitation, sunshine and the state of the ground. Different sports are tolerant to different ranges of each of the meteorological variables, and with this in mind we can classify the effect of weather on outdoor sports into three sections:

(1) *Specialised weather sports*

These sports require certain weather conditions in order to take place at all, and include sailing, gliding, and ski-ing requiring, respectively, wind velocity, thermals and snow. In general individual participants are competing against nature as well as against each other, and indeed nature may be sufficiently challenging to provide the only competition, as with *Crossbow* mentioned earlier. The specific weather requirements for these sports are fairly well documented. Recent publications include: for gliding – Wallington (1968), Wickham (1966); parachuting – D'Allenger (1970); hot-air ballooning –Samuel (1972); sailing – Watts (1967), Houghton (1969); and ski-ing – Perry (1972).

(2) *Weather Interference Sports*

Sports within this category are ideally suited to 'weatherless' days, for instance the weather should be warm, dry, bright but overcast, with little or no wind, excellent visibility, not too humid and with a firm ground surface, in fact, conditions in which most of the sports in the first category could not take place! Many of the weather interference sports are played on grass,

* For definition of geoclimate see footnote on p. 259

such as lawn tennis, soccer, rugby, hockey and bowls. Apart from lawn tennis these are team games which require a large playing area and hence usually take place outdoors. In most team games both sides are competing against each other at the same time, and are therefore experiencing the *same* weather conditions. These games usually incorporate two halves to compensate factors such as wind direction as well as the slope of the pitch. This subsection also includes individual sports such as running and walking events in athletics, and racing events such as cycling. Although some individuals may prefer adverse conditions, in general all competitors have to adjust to the same weather conditions. Weather principally acts as an equal interference factor, preventing play altogether in certain circumstances (such as thick fog).

(3) *Weather Advantage Sports*

a. Weather advantage – unequal interference

A qualifying golf tournament serves as an example of this; because of the numbers involved, not everyone can tee off at the first hole at the same time. Thus players who started out on a fine clear morning would have a clear advantage over players struggling to make par in a strong wind and pouring rain later in the day. Should the par score be altered according to the weather conditions? Any sport in which the 'arena' is too small to enable all competitors to compete at the same time, such as an athletic field event, is open to the possibility of a change in the weather if it lasts longer than the time scale of weather change. The Amateur Athletic Association (1976) requires that for records, information on the wind conditions must be available for all events up to 200 m and for the long jump and triple jump. The component of wind speed in the direction of motion is recorded at a height of four feet. If the component of the wind measured in the direction of running averages more than 2 m s^{-1} the record will not be accepted. The periods for which the wind component shall be measured are as follows:

All distances up to 100 m	10 s
120 yds or 110 m hurdles	15 s
220 yds and 200 m (including hurdles) straight	20 s
Long jump and triple jump	5 s

This, of course, only applies to the setting of records, although weather conditions may well provide unequal interference in running events if heats have to be run in which fastest losers qualify. The first heat may be run in sunshine and the last in a thunderstorm!

b. Unequal weather advantage in dual role sports

In certain sports, at any one time the participants or sides are performing different functions. The prime example is cricket in which one side is batting and the other bowling, which can lead to one side having a distinct advantage over the other due to the weather. In cricket a rain-affected pitch gives a distinct advantage to the bowling side. The weather can also act as an interference factor, of course, and this aspect is discussed in more detail elsewhere (Thornes 1976).

Few sports fall into this category because it requires not only that the sport lasts longer than the time required for weather change, but also, usually, that the weather change occurs when play is not taking place. First class cricket can last for either 3 or 5 days, in which time the weather is almost certain to change, but because play only lasts for about 6 hrs a day, changes are quite likely to take place between playing hours.

Baseball is somewhat similar to cricket in this respect. Striking and fielding are the two different roles, but it only lasts for a couple of hours at the most which is usually too short for dramatic weather change.

THE PLAYING SURFACE

The influence of the state of the playing surface for outdoor sports depends upon the sport considered. Most outdoor sports are played on grass, which is very weather sensitive. The soils and grass type can be of secondary importance if the soil storage capacity is full. Precipitation intensity, duration and timing will determine whether the surface is muddy, frosty or dry. Some sports are more tolerant than others to the state of the ground. Rugby for instance is far more tolerant than bowls, although on the other hand, a bowling green will undoubtedly be in use for a much greater period of time. The Sports Council in a recent publication state 'As a rule of thumb it is widely held that no more than three matches of adult football should be played on a normal grass pitch in a week; with more than that the surface may deteriorate rapidly'. Thus the land required for soccer or rugby pitches is wasted for much of the time, being used in season only about 4·5 hrs out of 168 hrs in a week. This has led to the development of artificial grass pitches which, it is claimed, can take up to 35 games a week. However a survey by the Sports Council found that 70 per cent of 104 soccer players interviewed thought that the style of play was different from that on an ordinary grass pitch. The artificial grass pitch when wet was thought to cause the ball to skid, but the playing surface was considered truer than a grass pitch (see Table 2). When dry it is considered to be an advantage for ball control but a disadvantage due to burns and bruises principally when slide tackling. Note that the weather is not defeated completely in that a thick fog can still halt play!

PARTICIPANT COMFORT

We have already mentioned, when considering Table 1, that the weather conditions, be they indoor or outdoor, are very important in determining participant comfort. Most sports involve a high activity level which means that the body is constantly generating excess heat as a by-product of inefficient energy conversion by the muscles: conditions must be such that stress does not result. Exercise changes the type of heat transfer within the body

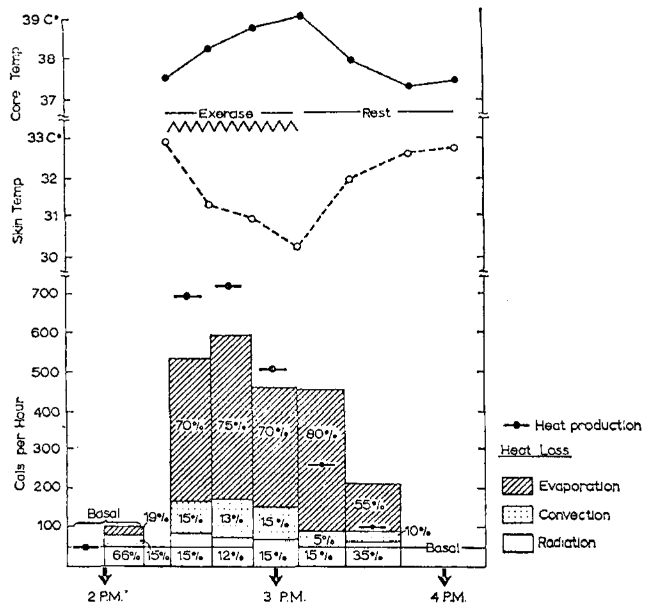

Fig. 2. Heat production and heat loss in a man playing squash (from Hardy *et al.*, 1938). Both participants were good players, evenly matched and they played almost to the point of exhaustion. The radiating temperature of the squash court wall averaged 21 °C and the air temperature averaged 20·5 °C. The skin temperature and the body radiation were averaged from 20 different places on the body surface. The players wore only light running trunks

and at the skin surface, in that body movements enhance heat loss via convection and evaporation. Fig. 2 shows the effect of core temperature (the deep body temperature), and skin temperature on the different proportions of heat loss for a typical game of squash. The core temperature rose steadily during the game, which is advantageous in that energy conversion by muscles is more efficient at higher temperatures (thus the importance of the warm-up before a game or race). Due to the increase in blood flow, exercise provides tolerance to much higher core temperatures than a resting individual can take without stress. The skin temperature falls because of the loss of latent heat via sweating and because of the convective heat loss. The sweat rate depends on the amount of movement, wind and the humidity of the air; the extent of convective loss depends on the amount of movement, wind, and the temperature of the air. Liquid sweat is no use as a cooling agent unless it evaporates. The ideal weather conditions will be different for each sport as far as participant comfort is concerned, because, of course, different sports have different activity levels. Table 3 lists typical amounts of heat produced per hour for various sports. Experiments in the Harvard Fatigue Laboratory (Morehouse and Miller 1953) have shown that the average individual cannot expend greater than 700 Kcal hr^{-1} for more than an hour. However marathon

TABLE 2. The advantages and disadvantages of artificial grass (Astroturf) as compared with normal grass football pitch

Item	Wet		Dry	
	Advantages	Disadvantages	Advantages	Disadvantages
Ball control	49	31	70	9
Stopping and turning	9	17	17	5
Tackling and falling	33	31	6	76
Others	9	21	7	10
	100	100	100	100

A sample of 104 soccer players, who regularly used an artificial pitch, were asked by the Sports Council to mention up to three advantages and three disadvantages of Astroturf under wet and dry conditions. The relative significance of each item therefore is reflected in the percentage of 'references', but the figures are not a direct percentage of players responding. The major disadvantage of dry Astroturf is due to the skin burns and bruising resulting from tackling and falling. For this reason it is often common practice to water the surface which leads to a reduction in ball control, principally due to the ball skidding

TABLE 3. Energy requirements of a man weighing 154 lbs undertaking various sporting activities. Units Kcal hr^{-1}

Billiards	235	
Baseball – normal	280	
– pitcher	390	
Table tennis	345	
Squash		600*
Fencing	630	
Wrestling	790	960*
American football	1000	800*
Rowing		1230*
Horizontal running 5 m s^{-1}	1300	
Cross country ski-ing		1400*
Crawl swimming 1·4 m s^{-1}		2000*
Horizontal running 8·5 m s^{-1}	9480	

From Morehouse and Miller (1953) except values marked * which are from Buskirk and Bass (1960)

Note that size, body type, age, physical fitness, skill, nutritional condition, environmental conditions etc. must all be taken into account for any one individual

runners averaging about 5 m s^{-1} expend at least 1300 Kcal hr^{-1} for more than 2 hrs. Water is a special environment since evaporation heat loss is severely curtailed, and most body cooling takes place via convection and conduction. Adee (1953) has shown that the optimum water temperature as far as performance is concerned is between 29 and 34·5°C for short races, and 23·0 and 26°C for 1500 m.

High activity sports require sparse clothing to enable freedom of movement. A balance must be sought between the required clothing and activity level for given environmental weather conditions. For indoor sports it should be possible for a given sport to define a maximum probable activity level and amount of clothing likely to be worn, in order to calculate the optimum temperature and humidity conditions for optimum performance. Different individuals may have different comfort zones, but dress can, of course, be changed accordingly.

In sports in which the hands play an important role, such as rugby, archery or hockey, low temperatures can be a distinct hazard. In cold conditions the blood to the hands and toes is reduced in far greater proportion than to the rest of the body (Bader and Mead 1950). This means that performance suffers as the sensitivity of the hands is lost.

Thus ecoclimatic and geoclimatic conditions are very important in determining the thermal comfort of a sportsman. In order for the sportsman to perform best he must be within his thermal comfort zone. Much remains to be established about the relationships, for different sports, between environmental conditions, activity level and required clothing.

SPECTATOR COMFORT

The spectator is a very important element in all professional sports, and even some amateur events, such as the Olympic Games. In general, adverse conditions will tend to attract the spectator to his television set rather than to watch an outdoor sport, or even an indoor sport if transport conditions are poor. Problems arise in that the spectator has usually to put up with exactly the same environmental conditions as the participants, and yet his activity level is much lower. Even for indoor sports conditions may be incompatible unless the spectator wears several layers of clothing, although he would be protected from the wind and the rain. The outdoor spectator is sometimes protected from the rain by stands but relies upon a packed crowd to provide wind protection.

The attendance at a sport depends not only upon spectator comfort but also upon the excitement and quality of the action. Presumably this is heightened if the player can give an optimum performance, and if the spectator can concentrate on the game, rather than on his cold feet! A compromise is possible for indoor sports in which the ecoclimate can be controlled. However it is impossible to control the geoclimate, and therefore the alternative is to improve spectator provisions. For instance at Houston in Texas a stadium to hold 60 000 spectators has been built with a roof over it, effectively transforming the outdoor sport of baseball into an indoor one. Maunder (1970) states:

'In 1964 . . . the year before the stadium was opened, the attendance was 725 000, compared with 2 539 000 in 1965, after the Astrodome was completed.'

The effect of weather on attendances is the most direct economic influence of weather on sport.

WEATHER ECONOMICS AND SPORT

Sport management must be constantly aware of the effect of weather upon its decisions. For instance Maunder mentions a certain August Saturday afternoon in New Zealand when the weather was so bad that 10 000 meat pies went unsold, because only half the expected 60 000 crowd turned up to watch an international rugby match.

In England the 1974 county cricket championship saw 46 days of cricket completely washed-out. To quote from Wisden (1975): 'The rain seeped into crickets profits . . . several counties found themselves in financial difficulties'. This was mainly due to the loss from spectator income, not only on the washed out days, but also in matches that were heavily interfered with by the weather, which took all the interest out of the match. In sports which last for only an hour or two a match can easily be rearranged if the weather prevents it taking place; thus if a soccer match is fog bound it is postponed and played at a later date. It is much more difficult to rearrange a county cricket match which lasts for 3 days, and hence all the potential revenue is lost.

CONCLUSION

This paper has attempted to classify the many inter-relations between weather and sport. Its subjective nature is due principally to the lack of research that has been undertaken in the whole field of the geography of sport (Rooney 1974). The possibilities for further research are considerable and by increasing the weather awareness of sport management not only could performances be improved, but also significant financial savings could be made.

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FOG-DEVIL

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THE photograph (Fig. 1) shows numerous small wisps of fog rising from an area where sunlight is falling on dew-covered ground and giving rise to a steaming effect, like Arctic sea smoke. In the centre of the patch of sunlight there is seen a pronounced vortex, reminiscent of a dust-devil. Two tall wisps to the right are the remains of an earlier vortex. In the left foreground there are patches of 'steam' beginning to rise at the edge of a new patch of sunlight.

At the time the picture was taken the sense of rotation of the vortex appeared to be anticyclonic, but it was difficult to be sure at the distance involved. The vertical component of the motion was clearly visible. The higher visible part of the vortex (not apparent in the photograph) appeared to reach about 250 feet above the base, i.e. to about the level of the hill-top behind.

The photograph was taken at about 1000 local time on Saturday 28 August 1976, looking in an east-north-easterly direction from a point about five miles south of Cooma, NSW, Australia. Earlier there had been a frost and super-cooled fog. The fog thinned between 0900 and 0930, when it started to break up and rise. A patch of sunlight broke through about 0945 and progressed slowly from left to right across the area shown, becoming diffuse 5 mins after the picture was taken. During this period its total movement was about one half-mile, and it was typically 500 yards across. Very soon after the sunlit patch appeared 'steam' began to rise from the ground, and shortly after that a vortex formed a little to the left of its centre. This moved slowly across the sunlit area until it broke up towards the right-hand edge, lasting 6 or 7 mins. It was followed by a second and a third vortex, the second being the largest in areal extent, and the third (the one pictured) the most intense.

There were still patches of frost on the grass at the time of the picture, and considerable melted frost or dew. The ground cover was mostly fairly short, dried grass, though there was some ploughed ground (dark areas in the photograph). There was no surface water around except for the creek whose course is indicated in the picture by the willow trees. The greyish strip seen in the foreground was a bitumen road.

The wind earlier had been very light and variable, but at the time of the picture a light breeze was becoming established from the north-east, i.e. blowing towards the camera and slightly from the left.

The photograph was taken with a Zeiss Ikon Continamatic camera, using a Kodacolour II film – it was in fact the 21st frame on a 20-exposure